

STELLA: Solar Temporal Event Learning & Likelihood Assessment

Project Proposal for UCS503P
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1 Higher-order goal

...secondary framing

This project aims to improve space-weather preparedness by reducing the time between the emergence of a significant solar event and an actionable warning. While solar flare forecasting is a broader scientific problem, the immediate engineering objective is to translate solar X-ray telemetry into a measurable and deployable software system for early detection, short-horizon forecasting, and impact assessment.

STELLA (Solar Temporal Event Learning & Likelihood Assessment) combines temporal solar observations, machine-learning-based event detection, forecasting, and explainable risk assessment into one monitoring workflow. Rather than only identifying a flare after it has developed, the system is designed to determine whether an event is emerging, estimate its likelihood and severity, and communicate its potential operational relevance.

2 Time-to-value

...primary heuristic

We will deliver meaningful increments quickly by first establishing a working solar telemetry pipeline and visualization layer, followed by a baseline flare detector. Forecasting, explainability, and impact assessment will then be added incrementally.

We will prioritize fast feedback over premature model optimization by validating each stage independently on historical solar observations before integrating the complete pipeline. The separation of nowcasting and forecasting also allows the first iteration to provide useful detection capability without waiting for the complete predictive system.

Success is defined by what can be delivered and measured in the first iteration: reliable telemetry processing, meaningful flare detection, measurable warning lead time, and an interpretable user-facing workflow.

3 Problem Statement

Solar flares are sudden releases of energy from the Sun that produce enhanced electromagnetic radiation and can contribute to disruptions in radio communication, navigation, satellite operations, and other space-dependent infrastructure. The engineering challenge is not simply to identify a flare after it occurs, but to recognize its emergence early enough to support preparation and response.

Several practical limitations make this difficult. Solar observations from different instruments capture complementary physical characteristics, but are often analyzed independently. Aditya-L1 provides valuable mission-specific observations, yet its available history is much smaller than long-running solar observation archives. Furthermore, solar background activity varies considerably over time, making fixed thresholds prone to false alarms. Finally, a prediction is less useful operationally when users cannot understand why it was generated or what systems may be affected.

Why this matters now (contextual relevance): Increasing dependence on satellite communication, navigation, remote sensing, and space-based infrastructure makes timely space-weather intelligence increasingly valuable. A system that turns raw solar observations into understandable warnings can reduce the gap between scientific monitoring and operational decision-making.

4 Proposed Solution

... *Software Proposition*

4.1 Overview

Build a real-time solar intelligence system that combines complementary X-ray observations associated with Aditya-L1 with temporal machine learning and operational visualization. STELLA will use SoLEXS soft X-ray observations and HELIOS hard X-ray observations to capture complementary thermal and higher-energy signatures of solar activity.

The system will perform telemetry ingestion, preprocessing, adaptive anomaly detection, temporal feature extraction, flare nowcasting, short-horizon forecasting, explainable prediction, and impact visualization. The final interface will present the current solar state, event likelihood, predicted flare severity, confidence, warning lead time, and potential infrastructure risk.

4.2 Core workflow

The core workflow is intentionally simple:

1. Solar X-ray telemetry is ingested and synchronized.
2. Temporal and spectral features are extracted and unusual activity is identified using adaptive statistics.
3. A Conv1D model performs short-window flare nowcasting.
4. A dilated Temporal Convolutional Network evaluates longer temporal context and estimates future event likelihood and severity.
5. The system generates an explanation and maps the prediction to potential infrastructure risks before presenting an alert through the dashboard.

This creates a complete operational chain:

Telemetry → Detection → Forecast → Explanation → Impact → Alert

4.3 Operational constraints

The system is designed as an engineering prototype rather than a replacement for operational space-weather agencies. Historical solar observations will be used for training and validation where mission-specific data is limited, while historical NOAA GOES observations can provide a larger training archive for transfer learning.

Where continuous real-time telemetry is unavailable, historical replay will reproduce the chronological behavior of an operational stream. Infrastructure outputs will be presented as modeled risk indicators rather than deterministic predictions of physical damage. The architecture will also remain modular so that additional solar instruments and data products can be incorporated later.

5 Solution Approach

... *Engineering focus*

This is an engineering course project, so the focus is on building a measurable, deployable temporal intelligence pipeline rather than claiming novel fundamental solar-physics research.

5.1 Temporal feature and anomaly assistance

... *physics-informed*

STELLA will derive features including soft and hard X-ray flux, flux rise rate, spectral hardness, rolling statistics, adaptive z-scores, and cross-instrument temporal relationships. A key physics-informed feature is the relationship between hard X-ray emission and subsequent soft X-ray evolution, motivated by the Neupert effect. This relationship will be used as an additional predictive signal rather than as a deterministic forecasting rule.

To handle changing solar background activity, the system will use rolling median absolute deviation (MAD):

$$MAD = \text{median}(|x_i - \text{median}(x)|)$$

and an adaptive standardized score:

$$Z_{\text{adaptive}} = \frac{x - \text{median}(x)}{1.4826 \times MAD + \epsilon}$$

This allows anomaly detection to adapt to changing background conditions rather than relying on a single global threshold.

5.2 Two-stage temporal architecture

... model design

The prediction pipeline separates immediate detection from longer-horizon forecasting. The **Conv1D nowcaster** operates on a short temporal window and estimates whether significant flare-like activity is emerging. The **dilated TCN forecaster** then uses a longer temporal context to estimate event likelihood, severity, and expected warning lead time.

This cascade is intended to keep the two objectives distinct: the first stage is optimized for rapid recognition of emerging signatures, while the second captures longer temporal dependencies required for forecasting.

5.3 Transfer learning

... limited-data strategy

Because mission-specific training data is limited, historical NOAA GOES X-ray observations will be used to learn generalized temporal representations. The model will first be trained on the larger historical archive, then adapted using available Aditya-L1 observations and evaluated on held-out events.

Chronological dataset splits will be preferred to random splits so that future observations do not inadvertently influence training and evaluation. This also makes the validation procedure more representative of the intended forecasting scenario.

5.4 Web architecture

... deployable solution

The system will use a modular web architecture. The frontend will provide the monitoring dashboard, while a FastAPI backend will handle telemetry ingestion, inference requests, historical data, alerts, and system status. Python-based inference services will execute the feature-processing and temporal models, with REST APIs and WebSockets supporting standard requests and live updates.

The architecture separates frontend, backend, data processing, and inference so that individual components can be tested and scaled independently.

5.5 CI/CD and fast delivery enabler

CI/CD will automatically build the application, run unit and integration tests, validate API contracts and inference sanity checks, and produce repeatable staging deployments. This supports frequent incremental releases while reducing the risk of breaking the complete monitoring pipeline during model or interface changes.

6 Evaluation Criterion

... measurable and attributable

We define success using measurable metrics tied directly to the forecasting workflow. The primary metric is **warning lead time**: the time between the first valid system warning and the defined reference time for the corresponding solar event.

6.1 Primary evaluation metric

Warning lead time: The initial target is at least 15 minutes of useful warning for qualifying events, with approximately 30 minutes as a stretch objective. The value will be calculated using timestamped telemetry, model predictions, and historical flare-event catalogues.

6.2 Secondary evaluation metrics

- **Probability of Detection (POD):** fraction of qualifying events correctly identified.
- **False Alarm Ratio (FAR):** fraction of generated warnings that do not correspond to qualifying events.
- **Critical Success Index (CSI):** combined measure of hits, misses, and false alarms.
- **Inference latency:** time required to convert new telemetry into a prediction.
- **Forecast calibration:** agreement between predicted likelihoods and observed event frequencies.

6.3 Pilot validation plan

... high-level

Validation will proceed from offline evaluation to operational replay. First, models will be evaluated on chronologically held-out historical events. Historical telemetry will then be streamed through the complete pipeline to simulate real-time operation. Finally, the deployed system will be evaluated for ingestion latency, inference latency, warning generation, false alarms, and dashboard responsiveness.

7 Scalability

... with foresight

The proposition should scale in both theoretical and practical senses.

1. **Scalable (theoretically):** The system can support increasing telemetry volume and additional instruments through decoupled frontend, backend, and inference services, stateless APIs, asynchronous processing, and indexed historical data.
2. **Operationally deployable (practically):** The system can run on common cloud or local infrastructure using containerized services, managed or local databases, standard Python ML environments, WebSocket-enabled APIs, and CI/CD pipelines.

The modular design also allows new instruments or derived solar data products to be introduced without redesigning the complete application.

8 Engine availability heuristic

A mature set of “engines” is available to implement this as a solar intelligence project. Historical solar X-ray archives provide training and validation data, while Python-based scientific and machine-learning frameworks support temporal feature processing and model development. Conv1D and Temporal Convolutional Network architectures provide established mechanisms for temporal learning.

FastAPI can provide REST and WebSocket services, while React/Vite can support the monitoring interface. Standard testing frameworks and CI runners can provide repeatable validation and deployment.

We will use these mature components to focus on temporal feature fusion, forecasting, explainability, system integration, and measurement rather than inventing new infrastructure.

9 Project scope and deliverables

... iteration-friendly

9.1 Initial deliverable

...first iteration

The first iteration will establish the minimum end-to-end system:

- Solar telemetry ingestion and preprocessing.
- Historical GOES data pipeline and temporal feature extraction.
- Adaptive MAD-based anomaly detection.
- Initial Conv1D nowcasting model.
- Basic inference API and telemetry visualization.
- Prediction timestamp logging, CI testing, and staging deployment.

The objective is to demonstrate a complete working path from telemetry to an interpretable detection output before adding the more advanced forecasting and impact layers.

9.2 Subsequent deliverables

Subsequent iterations will add the dilated TCN forecasting model, SoLEXS/HEL10S multi-instrument fusion, GOES-to-Aditya-L1 transfer learning, lead-time estimation, explainable AI, historical event catalogues, and the India-specific impact layer.

The final dashboard will combine live telemetry, event likelihood, predicted class, confidence, lead time, explanation, alerts, and infrastructure risk into a single operational view.

10 Risks and mitigations

- **Limited mission-specific data:** use historical GOES observations for pre-training and adapt using available Aditya-L1 data.
- **Instrument domain shift:** apply instrument-specific normalization and validate individual signals before fusion.
- **False alarms:** use adaptive thresholds and explicitly measure FAR under different solar activity levels.
- **Data leakage:** use chronological splits and prevent observations from the same event from crossing training and testing partitions.
- **Limited real-time telemetry:** maintain a chronological historical replay mode for end-to-end testing.
- **Model overfitting:** compare against simple baselines and evaluate on held-out events.
- **Impact uncertainty:** communicate infrastructure results as modeled risk indicators rather than deterministic damage predictions.

11 Summary

This project proposes a deployable solar intelligence platform that converts temporal X-ray observations into early-warning information for significant solar events. It emphasizes time-to-value through rapid iterations, CI/CD for reliable releases, and measurable evaluation through warning lead time, detection performance, false alarms, and inference latency.

STELLA combines multi-instrument observations, adaptive anomaly detection, Conv1D nowcasting, TCN forecasting, transfer learning, explainable AI, and impact visualization within a single workflow. The system remains focused on engineering a reliable decision-support pipeline rather than claiming novel solar-physics discovery.

Its modular architecture supports future instruments and data products, while historical replay enables meaningful validation even when continuous real-time telemetry is unavailable. The final system provides a unified interface to observe solar activity, identify emerging events, estimate their likelihood, understand prediction drivers, and assess potential infrastructure relevance.